Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

## COURSE: ADAPTIVE SOFTWARE ENGINEERING– (24CI3201)
## A.Y. 2026 - 2027

| **TITLE OF THE PROJECT** | **AI-Driven Adaptive Software Quality Assurance and Security Automation Platform** | |
|---|---|---|
| **S. No.** | **University ID** | **Name** |
| 1 | 2420030540 | Allipuram Akhil |
| 2 | 2420030546 | P. Ruthvik Reddy |
| 3 | 2420030579 | S. Akshay Veer |
| 4 | 2420030141 | B. Akhil Varma |
| Name of the Guide | | Dr.J.Kavitha Reddy |


## Abstract

Modern software development requires effective quality assurance and security practices as applications become increasingly complex and continuously updated through CI/CD pipelines. Traditional testing and security approaches are often manual, static, and disconnected, making it difficult to efficiently identify defects and vulnerabilities. This project proposes an **AI-Driven Adaptive Software Quality Assurance and Security Automation Platform (ASSQP)** that integrates automated testing, **DevOps and CI/CD practices**, **static code analysis using SonarQube**, and **security scanning through DevSecOps tools such as OWASP ZAP and Trivy** into a unified framework. The platform analyzes source code, test results, logs, code changes, and vulnerability data to identify quality and security issues. An ML-based risk prediction model uses factors such as code complexity, code churn, historical defects, vulnerability severity, and test coverage to classify software components according to their risk levels. Based on these predictions, the system implements adaptive test prioritization, where high-risk components receive more extensive testing while lower-risk components undergo essential tests, improving testing efficiency and resource utilization. The application is containerized using Docker and integrated with a CI/CD pipeline for automated building, testing, security scanning, and deployment. The system also follows **MLOps principles** for data collection, model training, versioning, deployment, and monitoring. Finally, a centralized dashboard presents defects, vulnerabilities, test coverage, risk scores, affected components, and recommended corrective actions. By combining DevOps, SecOps, MLOps, machine learning, and adaptive software engineering, ASSQP aims to provide a continuous and feedback-driven approach that improves software reliability, security, and testing efficiency.


Signature of the Guide HOD